

SENTINELT

Robo Wars Combat Robot | Robo Rumble 2026 Elimination Round

SentinelT

Compact RC Combat Robot Concept

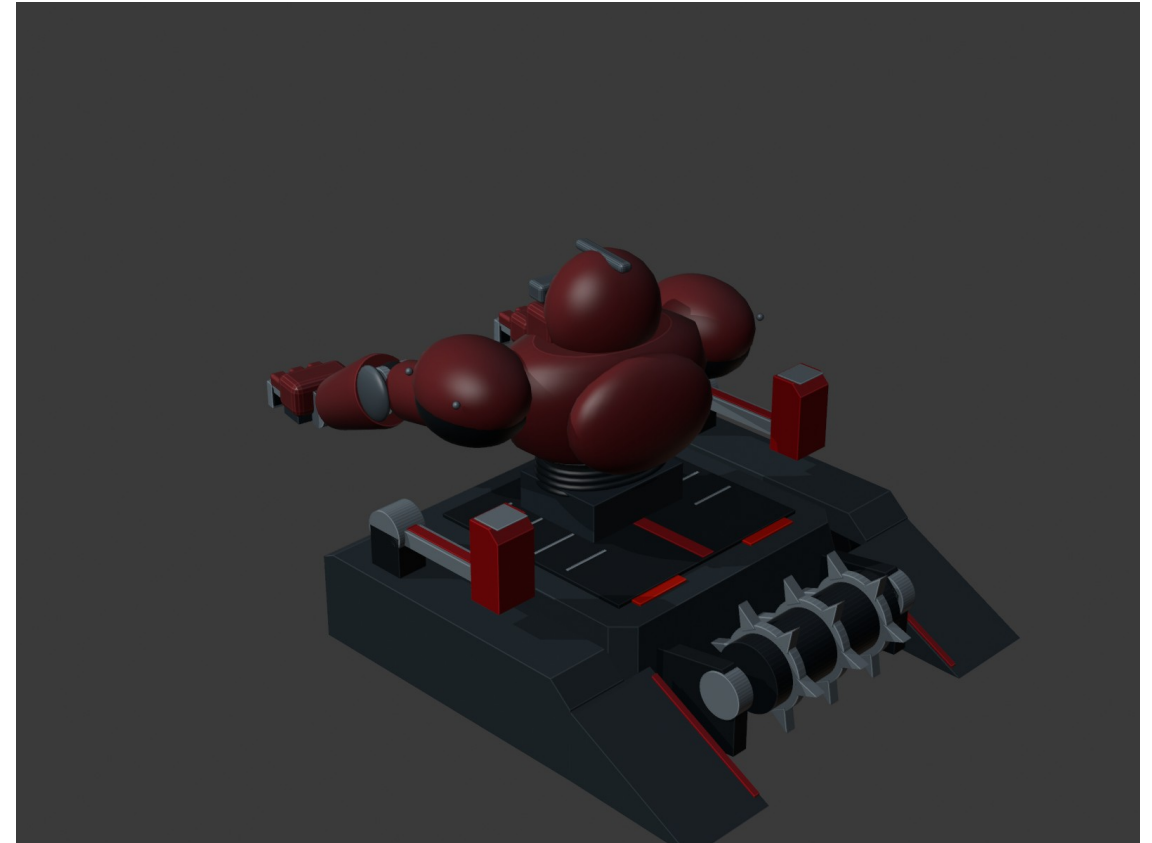
Designer: Thato Glen Assegai

Category: Robo Wars - RC Combat

Compliant CAD footprint and engineering mass budget

318.16 x 495.00 mm footprint

180 / 180 frames PASS



1. Problem & Competition Constraints

Designing a combat robot inside strict rules

Maximum footprint: 500 mm x 500 mm

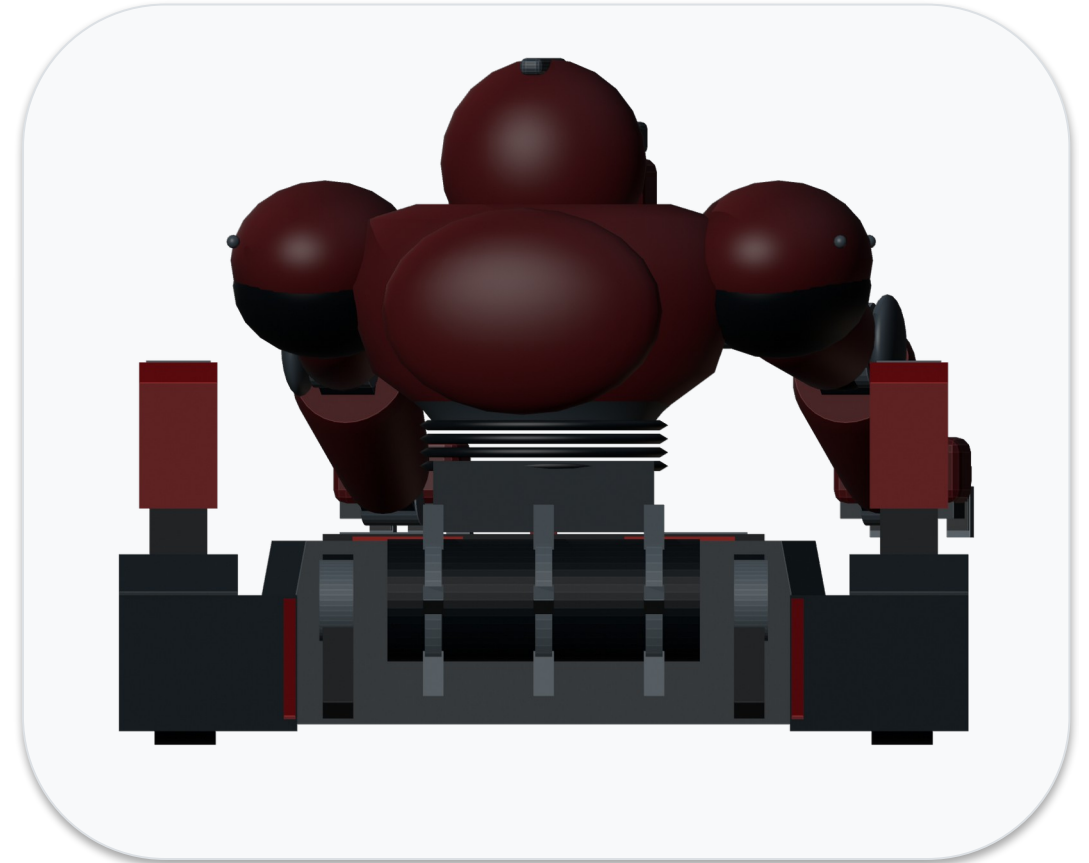
Physical mass: strictly below 5 kg

Human-operated control: 2.4 GHz RC only

Mandatory accessible ON/OFF switch

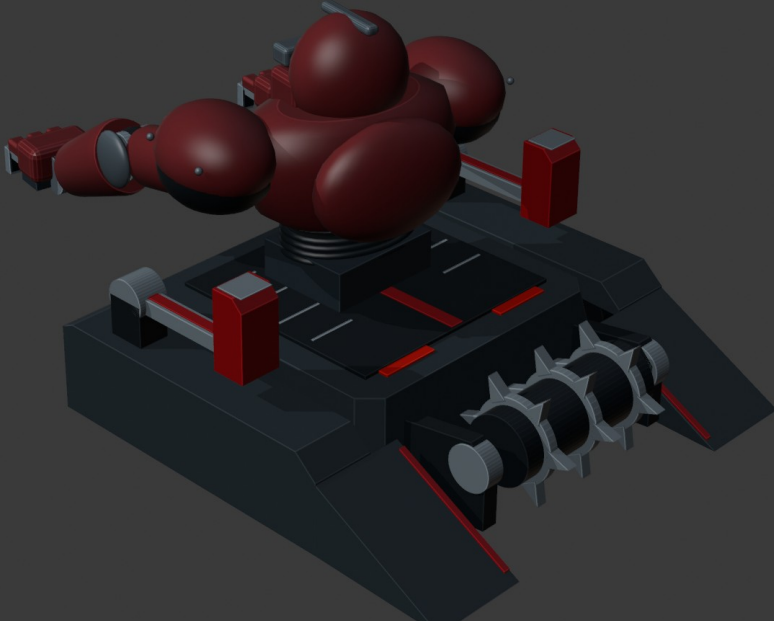
Mandatory Emergency Stop (E-Stop)

No projectiles, flames or liquids



2. SentinelT Solution

Low-profile armored platform with safety-focused RC architecture



Armored mobile chassis

Protected wheel and drive areas

Sloped frontal and side protection

Articulated upper structure

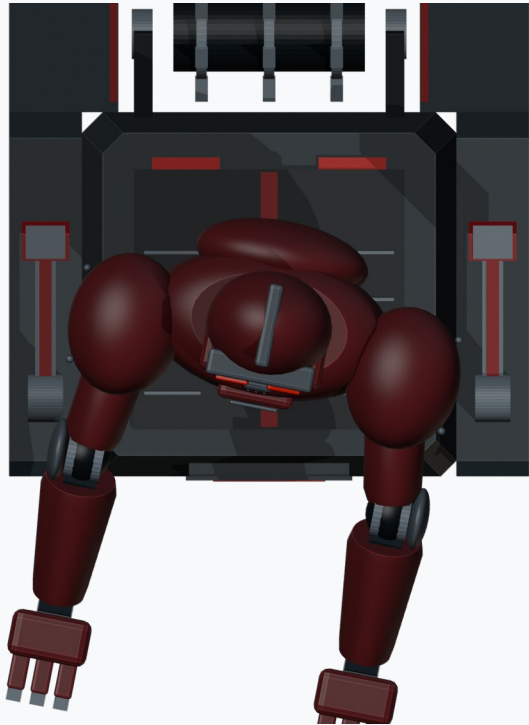
Serviceable and removable covers

Guarded active-mechanism provision

2.4 GHz human-operated RC

3. CAD Compliance & Simulation

Blender/Python audits verified dimensional compliance



WIDTH: 318.16 mm

LENGTH: 495.00 mm

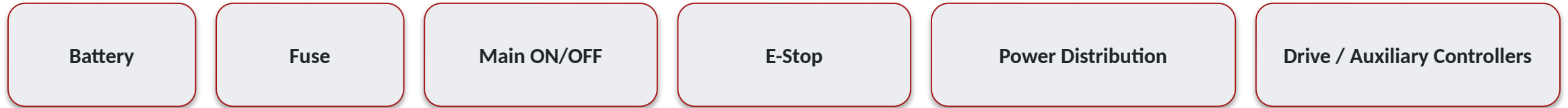
HEIGHT: 255.41 mm

CHECKED FRAMES: 180 / 180 PASS

Footprint status: PASS

4. Electronics, 2.4 GHz Control & Safety

Power isolation and E-Stop are explicit design requirements



2.4 GHz RC receiver -> controller -> drive commands

Loss of RC signal / invalid input -> safe-state actuator disable

E-Stop is treated as a hardware safety layer

Final current and voltage ratings must be verified on selected components

5. Buildability, Mass & Cost

Engineering allocation is compliant and procurement is in progress

Engineering mass budget: 4.650 kg

Remaining margin to 5 kg: 0.350 kg

Planned fabrication: plate/sheet structure, bolted assemblies and serviceable covers

Current project status: design and documentation stage

BOM spreadsheet included

VERIFIED CANDIDATE SUBTOTAL

R 3,622.15

Includes RC set, dual brushed ESC, battery, E-Stop, fuse holder and XT60 connector.

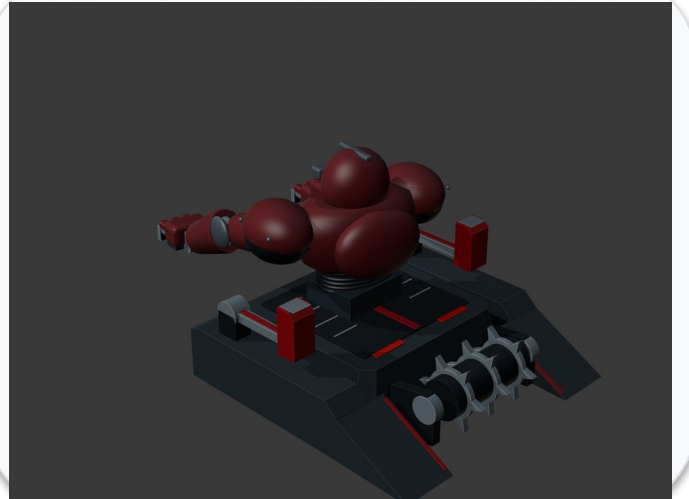
Final implementation total still depends on wheels, chassis/armor, fasteners, wiring and final active-mechanism hardware.

6. Individual Entrant & Responsibilities

Thato Glen Assegaai - SentinelIT designer



Concept development
CAD modelling and Blender scripting
Mechanical architecture
2.4 GHz control framework
ON/OFF and E-Stop safety architecture
Documentation and GitHub repository



Solo entrant profile updated with supplied photo. This white-theme version is ready for submission review.